

ST AUGUSTINE AP, FL, USA

WMO: 722212

Lat: 29.959N

Lon: 81.340W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 92814

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.1	2.9	-7.8	1.9	5.8	-5.0	2.5	6.8	11.1	12.4	9.8	12.1	2.9	310	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	33.6	25.4	32.5	25.5	31.5	25.4	27.0	31.0	26.6	30.7	26.2	30.1	4.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.3	29.5	25.3	20.5	29.1	25.0	20.1	28.7	84.7	31.2	82.6	30.6	81.0	30.1	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.4	7.6	-2.6	35.5	1.5	1.3	-3.7	36.5	-4.6	37.2	-5.5	38.0	-6.6	38.9		
			-4.1	27.9	1.9	0.7	-5.5	28.5	-6.6	28.9	-7.7	29.3	-9.0	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	13.2	15.0	17.2	20.5	23.9	26.6	27.5	27.5	26.3	22.7	18.2	15.0
	DBStd	6.13	5.06	4.51	4.20	3.09	2.41	1.61	1.28	1.24	1.50	3.32	4.12	4.91
	HDD10.0	54	28	9	3	0	0	0	0	0	0	0	2	12
	HDD18.3	552	171	111	70	15	1	0	0	0	0	7	52	125
	CDD10.0	4132	126	148	227	314	431	499	542	542	491	395	248	168
	CDD18.3	1588	10	17	36	79	173	249	284	283	241	144	48	23
	CDH23.3	13192	34	67	164	416	1269	2249	2780	2803	2195	971	176	69
(13)	CDH26.7	4038	2	7	24	66	318	780	1054	997	611	160	17	4
(14)	Wind	WSAvg	3.6	3.5	3.6	3.9	4.0	4.0	3.5	3.2	3.9	3.9	3.7	3.4
(15) Precipitation	PrecAvg	1292	72	72	87	63	75	174	138	163	219	122	56	64
	PrecMax	1682	201	217	225	224	180	284	292	327	433	306	144	257
	PrecMin	835	24	16	4	14	26	18	42	73	51	8	4	10
	PrecStd	225	47	52	60	53	48	67	59	65	113	78	36	64
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.4	27.7	29.2	30.9	33.3	35.4	35.1	34.3	32.8	31.1	28.3	27.2
		MCWB	19.9	20.2	20.3	21.5	23.0	25.1	25.7	25.8	24.6	25.0	23.2	22.3
	2%	DB	23.9	25.5	27.0	28.3	31.2	33.3	33.1	32.8	31.3	29.3	27.0	25.5
		MCWB	19.3	19.9	19.4	21.3	22.7	25.1	25.7	26.1	25.0	24.3	22.4	21.3
	5%	DB	22.3	23.4	25.2	27.1	29.4	31.9	32.3	32.0	30.4	28.3	25.3	23.3
		MCWB	18.4	18.7	19.2	20.8	23.0	25.2	25.8	26.2	25.0	23.6	21.4	20.3
	10%	DB	20.6	21.8	23.5	25.9	28.1	30.6	31.2	31.0	29.6	27.4	24.0	22.0
		MCWB	17.6	18.1	18.4	20.3	22.6	24.9	25.6	25.9	24.8	22.9	20.5	19.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.6	21.9	22.5	23.9	25.3	27.0	27.5	27.7	26.9	26.2	24.7	23.2
		MCDB	24.3	25.1	26.0	27.7	29.6	31.5	31.7	31.6	30.2	29.2	27.1	25.8
	2%	WB	20.3	20.8	21.2	23.0	24.6	26.3	26.9	27.1	26.2	25.3	23.6	22.0
		MCDB	22.9	23.9	24.8	26.5	28.5	31.2	31.2	31.0	29.6	28.2	25.8	24.4
	5%	WB	19.2	19.8	20.4	22.2	24.0	25.7	26.5	26.7	25.7	24.6	22.4	21.0
		MCDB	21.7	22.5	23.8	25.5	27.7	30.2	30.8	30.6	29.0	27.4	24.5	22.9
	10%	WB	18.0	18.6	19.5	21.4	23.4	25.3	25.9	26.2	25.3	24.0	21.2	19.9
		MCDB	19.9	21.0	22.4	24.6	27.1	29.4	30.1	30.0	28.6	26.7	23.5	21.7
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	9.2	9.4	9.1	8.3	7.8	8.0	7.3	6.4	7.6	8.7	9.1
		MCDBR	10.4	10.6	10.8	10.0	10.0	10.0	9.3	8.6	7.5	7.3	8.5	9.2
		MCWBR	6.8	6.1	5.8	5.1	4.2	4.1	4.0	3.8	3.4	3.7	5.0	5.9
	5% WB	MCDBR	9.3	9.4	9.2	8.5	8.1	8.5	8.3	7.6	6.6	6.3	7.1	8.4
		MCWBR	6.5	6.0	5.7	4.8	4.0	4.1	4.1	3.9	3.4	3.4	4.8	5.7
(40) Clear-Sky Solar Irradiance	taub		0.339	0.352	0.371	0.396	0.433	0.472	0.497	0.482	0.427	0.400	0.362	0.346
	taud		2.540	2.513	2.461	2.375	2.314	2.270	2.203	2.252	2.401	2.444	2.498	2.542
	Ebn at Noon		896	915	915	899	862	824	801	811	848	850	862	869
	Edh at Noon		87	97	109	123	131	136	145	137	114	103	90	83
(44) All-Sky Solar Radiation	RadAvg		3.16	3.80	4.96	6.05	6.42	5.98	5.93	5.55	4.79	4.24	3.42	2.87
	RadStd		0.23	0.39	0.43	0.36	0.41	0.46	0.37	0.36	0.39	0.35	0.20	0.25

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	+0.44	+0.77	N/A	N/A	N/A	N/A	-9	-62	+160	+116

Nomenclature: See separate page